MIL-A-85097/14B(AS)
28 SEPTEMBER 1988
SUPERSEDING
MIL-A-85097/14A(AS)
27 July 1987

MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET CATAPULT MARK 1 MOD 1 (LOADED ASSEMBLY)

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the Mark 1 Mod 1 Rocket Catapult described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to the Mk 1 Mod 1 Rocket Catapult (loaded assembly) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the cover specification, MIL-A-85097, and form a part of this specification sheet to the extent specified herein.

DRAWINGS

Naval Air Systems Command (CAGE Code 30003)

DL1516349	Container, Shipping and Storage, Mk 217 Mod 0
DL657428	Propellant Grain, Mk 60 Mod 0
DL4904030	Igniter, Rocket Motor, Mk 189 Mod 1
DL4904215	Igniter, Rocket Motor, Mk 250 Mod 1
4904288	Box, Shipping and Storage, Mk 1 Mod 1/Rocket Catapult
DL1517547	Firing Mechanism, Rocket, Mk 28 Mod 2 Assembly
DL1517580	Catapult, Aircraft Ejection Seat, Rocket, Mk 1 Mod 1, Loaded Assembly

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC: N/A

FSC 1377

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3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The loaded Rocket Catapult shall conform to the documents specified herein and the applicable documents listed on DL4904215, DL4904030, DL657428, DL1517547, and DL1517580.

3.2 Preproduction/production.

3.2.1 Ballistic performance. The rocket catapult shall meet the requirements of tables I and II when test fired using an ejected weight of 317 pounds (see 6.2 for definitions).

TABLE I. Static firing requirements and AQL's for booster.

Performance requirements	Firing temperatures and AQL percentages					
	−40 °F			160 °F		
	Max	Min	AQL %	Max	Min	AQL %
Ignition delay (ms)	30	0	1.0	30	0	1.0
Maximum acceleration (g)	11	—	0.65	20	—	1.0
Onset rate (g/s)	200	—	1.0	350	—	1.0
Separation velocity (ft/s)	—	30	1.0	—	45	1.0
Launch time (ms)	260	—	2.5	170	—	2.5

TABLE II. Static firing requirements and AQL's for sustainer.

Performance requirements	Firing temperatures and AQL percentages					
	−40 °F			160 °F		
	Max	Min	AQL %	Max	Min	AQL %
Ignition delay (ms)	30	0	2.5	10	0	2.5
Maximum resultant thrust (lb)	4220	2590	2.5	5500	3355	2.5
Impulse (lb-s)	—	1140	2.5	—	1140	2.5
Action time (ms)	555	410	2.5	400	325	2.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction.

4.1.1 Sample size. Twenty (20) rocket catapults shall be submitted for preproduction inspection and tested in accordance with table III of MIL-A-85097.

4.1.2 Sample inspection.

4.1.2.1 Propellant grain Mk 60 Mod 0. The propellant grain, prior to assembly into the loaded rocket catapult, shall meet the requirements of the documents listed on DL657428. The propellant grains shall be considered acceptable when the loaded rocket catapult is in compliance with 3.2.1.

4.1.2.2 Igniter Mk 250 Mod 1. Ballistic requirements of the igniter shall be considered acceptable when the loaded rocket catapult is in compliance with 3.2.1.

4.1.2.3 Igniter Mk 189 Mod 1. Ballistic requirements of the igniter shall be considered acceptable when the loaded rocket catapult is in compliance with 3.2.1.

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4.2 Production.

4.2.1 Sample size. Samples selected for production testing shall be as specified in table III.

TABLE III. Static firing tests.

Lot size	Sample size	Firing temperatures ±5 °F	
		–40 °F	160 °F
0 - 40	6	3	3
41 - 65	8	4	4
66 - 110	10	5	5
111 - 180	15	7	7
181 - 300	20	10	10

4.2.2 Sample inspection. The production test sample, when in the static firing arrangement, shall be considered acceptable when the loaded rocket catapults are in compliance with 3.2.1.

4.2.3 Conditioning period. Time for conditioning prior to static firing shall be not less than 4 hours.

5. PACKAGING

5.1 Packing. Exterior packing of two loaded Mk 1 Mod 1 Rocket Catapults shall be in accordance with DL1516349. Single unit packing shall be in accordance with Drawing 4904288.

6. NOTES

6.1 Intended use. The Mk 1 Mod 1 Rocket Catapult is intended for use in the A-4E/L aircraft for ejecting the pilot.

6.2 Definitions.

6.2.1 Booster motor.

6.2.1.1 Maximum acceleration. The maximum acceleration is defined as the highest acceleration during the booster burning cycle as scaled from the acceleration-time curve. Accelerations are to be expressed in g's.

6.2.1.2 Ignition delay. Ignition delay (ID) is defined as the elapsed time between break wire break and the first indication of catapult thrust.

6.2.1.3 Maximum onset rate. The maximum onset rate shall be measured from the acceleration curve over the 0.030 second interval that produces the greatest rate.

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6.2.1.4 Separation velocity. Separation velocity is defined as the average velocity of the simulated ejected weight propelled horizontally at catapult separation measured over the interval from 6 inches before to 6 inches after separation.

6.2.1.5 Launch time. Launch time is defined as the time between first indication of thrust on the axial thrust curve and the sudden drop in thrust at the end of the catapult phase on the thrust curve.

6.2.2 Sustainer motor.

6.2.2.1 Maximum resultant thrust. Maximum resultant thrust is defined as the maximum vectorial sum of all components of rocket motor thrust in any one instant.

6.2.2.2 Ignition delay. Ignition delay is the elapsed time from separation (the point on the catapult thrust trace where the thrust begins to drop sharply) to 10 percent of maximum thrust, as measured on the rising portion of the normal component rocket thrust trace.

6.2.2.3 Action time. Action time (t_A) is the time interval from the 10 percent of maximum thrust on the initial rise of the normal curve to the corresponding 10 percent of maximum thrust on the declining portion of the curve.

6.2.2.4 Impulse. The impulse is obtained by vectorially adding the longitudinal and normal impulse of the rocket motor taken over the action time.

6.2.2.5 Acceptable quality level (AQL). AQL is a nominal value expressed in terms of percent defective specified for a single quality characteristic.

6.3 Changes from previous issue. Revision letters are not used to denote changes due to the extensiveness of the changes.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-85097/14B(AS)	AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS): ROCKET CATAPULT MARK 1 MOD 1 (LOADED ASSEMBLY)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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